$$\text{Direct form: } Q^S(P), \quad Q^D(P) \tag{1}$$

$$\text{Inverse form: } P^S(Q), \quad P^D(Q) \tag{2}$$

$$Q^S = P - 50 \;\Leftrightarrow\; P = 50 + Q \tag{3}$$

$$Q^D = 85 - 0.5P \;\Leftrightarrow\; P = 170 - 2Q \tag{4}$$

- Valid for $P \geq 50$ (supply), $P \leq 170$ (demand)
- Inverse slope $= 1/$ direct slope
- Demand: -0.5 and -2; supply: 1 and 1

$$TC(q) = a + bq + cq^2 \tag{5}$$

$$MC(q) = \frac{dTC}{dq} = b + 2cq \tag{6}$$

- a: fixed cost, disappears in the derivative
- bq contributes b; cq^2 contributes $2cq$
- MC: the cost of one more unit

$$TVC(q) = \int_0^q MC(x)\,dx = bq + cq^2 \qquad (7)$$

$$PS = TR - TVC = P \times q - TVC(q) \qquad (8)$$

$$= \int_0^q \big[P - MC(x)\big]\,dx$$

- Integral = area under MC, 0 to q
- PS: between price line and MC

$$\text{Area of a triangle} = \tfrac{1}{2} \times \text{base} \times \text{height} \tag{9}$$

$$CS = \tfrac{1}{2}\, Q^* \,(\text{demand intercept} - P^*) \tag{10}$$

$$PS = \tfrac{1}{2}\, Q^* \,(P^* - \text{supply intercept}) \tag{11}$$

- Base $= Q^*$ on the quantity axis
- Height $=$ vertical distance on the price axis

$$\int_0^Q (A - bx)\,dx = AQ - \tfrac{1}{2}bQ^2 \tag{12}$$

$$\int_0^Q (u + sx)\,dx = uQ + \tfrac{1}{2}sQ^2 \tag{13}$$

$$220 \times 40 - \tfrac{1}{2} \times 40^2 = 8{,}800 - 800 = 8{,}000 \tag{14}$$

$$50 \times 40 + \tfrac{1}{2} \times 40^2 = 2{,}800 \tag{15}$$

- Under $P = 220 - Q$ and $MC = 50 + Q$
- Needed in Question 3.4d

$$TC_N(q) = 20 + 50q + 5q^2 \tag{16}$$

$$TC_B(q) = 20 + (P_N + 50)q + 5q^2 \tag{17}$$

$$P_B = 220 - Q_B, \qquad Q_N = Q_B \tag{18}$$

- q: one firm; Q: industry (10 firms)
- $P_N q$: the nickel bill
- Innovation: $TC_N = 20 + 20q + 5q^2$

$$P = MC(q) = 50 + 10q \tag{19}$$

$$\Rightarrow 10q = P - 50$$

$$\Rightarrow q(P) = \frac{P - 50}{10}, \quad P \geq 50$$

- Firm's supply curve = its MC curve
- Direct form: q as a function of P
- $P < 50$: no supply (first unit costs 50)

$$Q_N(P) = 10 \times q(P) = 10 \times \frac{P - 50}{10} \tag{20}$$
$$= P - 50$$

$$P_N = 50 + Q_N \tag{21}$$

- Horizontal sum: add quantities, not prices
- Slope: firm 10, industry 1
- A £1 rise: $+0.1$ per firm, $+1$ in total

$$MC(q) = \frac{dTC_N}{dq} = 20 + 10q \tag{22}$$

$$q(P) = \frac{P - 20}{10}, \quad P \geq 20 \tag{23}$$

$$Q_N = P - 20 \;\Leftrightarrow\; P_N = 20 + Q_N \tag{24}$$

- Down by 30 at every Q (the cost change)
- Right by 30 units (industry, slope 1)
- Right by 3 units (firm, slope 10)

$$P_N = 90: \quad q = \frac{90 - 50}{10} = 4 \tag{25}$$

$$Q_N = 10 \times 4 = 40 = 90 - 50 \checkmark \tag{26}$$

$$P_N = 120: \quad q = \frac{120 - 50}{10} = 7 \tag{27}$$

$$Q_N = 10 \times 7 = 70 = 120 - 50 \checkmark \tag{28}$$

- Industry $= 10\times$ one firm, at every price
- Industry curve: $Q_N = P_N - 50$

$$\text{After, } P_N = 90: \quad q = \frac{90 - 20}{10} = 7 \tag{29}$$

$$Q_N^S = 90 - 20 = 70 \quad (= 10 \times 7) \tag{30}$$

$$Q_N^D = 85 - 0.5 \times 90 = 40 \tag{31}$$

$$\text{Excess supply} = 70 - 40 = 30 \tag{32}$$

- Unsold nickel: sellers undercut each other
- The price falls until $Q_N^S = Q_N^D$

$$170 - 2Q = 50 + Q \tag{33}$$
$$\Rightarrow 3Q = 120$$
$$\Rightarrow Q_0 = 40$$

$$P_0 = 50 + 40 = 90 \tag{34}$$

Check: $170 - 2 \times 40 = 90$ ✓

- Inverse form: set the two prices equal

$$85 - 0.5P = P - 50 \tag{35}$$
$$\Rightarrow 1.5P = 135$$
$$\Rightarrow P_0 = 90$$

$$Q_0 = 90 - 50 = 40 \tag{36}$$

- Direct form: set the two quantities equal
- $E_0 = (Q_0, P_0) = (40, 90)$
- 4 units per firm, as in Exercise 1

$$CS_{N,0} = \tfrac{1}{2} \times 40 \times (170 - 90) \qquad (37)$$
$$= \tfrac{1}{2} \times 40 \times 80 = 1{,}600$$

$$PS_{N,0} = \tfrac{1}{2} \times 40 \times (90 - 50) \qquad (38)$$
$$= \tfrac{1}{2} \times 40 \times 40 = 800$$

$$TS_{N,0} = 1{,}600 + 800 = 2{,}400 \qquad (39)$$

- Check: 10 firms $\times$ 80 $=$ 800

$$\varepsilon_D = \frac{\%\Delta Q}{\%\Delta P} = \frac{\Delta Q/Q}{\Delta P/P} \tag{40}$$
$$= \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$\varepsilon_D = \frac{dQ}{dP} \times \frac{P}{Q} \qquad \text{(point elasticity)} \tag{41}$$

- dQ/dP: slope of the direct form $Q(P)$
- P/Q cancels the units: a pure number

$$\varepsilon_S = \frac{dQ^S}{dP} \times \frac{P}{Q} > 0 \tag{42}$$

$$b = \frac{P/Q}{|\varepsilon_D|}, \qquad s = \frac{P/Q}{\varepsilon_S} \tag{43}$$

- b, s: slopes of the inverse curves
- $|\varepsilon_D|$: elastic > 1, unit $= 1$, inelastic < 1
- Vertical demand: $\varepsilon_D = 0$ (b infinite)
- Horizontal demand: $\varepsilon_D \to -\infty$ ($b = 0$)

$$Q = 100 - 2P \Rightarrow \frac{dQ}{dP} = -2 \tag{44}$$

$$\varepsilon_D\big|_{P=20} = -2 \times \frac{20}{60} = -0.67 \tag{45}$$

$$\varepsilon_D\big|_{P=40} = -2 \times \frac{40}{20} = -4 \tag{46}$$

- $P = 20$: $Q = 60$ (inelastic)
- $P = 40$: $Q = 20$ (elastic)
- One slope, a new elasticity at every point

$$Q_N = 85 - 0.5P_N \Rightarrow \frac{dQ}{dP} = -0.5 \tag{47}$$

$$\varepsilon_D = -0.5 \times \frac{90}{40} = -1.125 \tag{48}$$

$$Q_N = P_N - 50 \Rightarrow \frac{dQ}{dP} = 1 \tag{49}$$

$$\varepsilon_S = 1 \times \frac{90}{40} = 2.25 \tag{50}$$

- Price $+1\%$: demand about -1.1%
- Price $+1\%$: supply about $+2.25\%$

$$\rho = \frac{\text{price fall}}{\text{cost fall}} = \frac{20}{30} = \frac{2}{3} \tag{51}$$

$$P = A - bQ, \qquad P = u + sQ \tag{52}$$

$$u \longrightarrow u - k \tag{53}$$

- $b, s \geq 0$: steepness of demand and supply
- Nickel: $A = 170$, $b = 2$, $u = 50$, $s = 1$
- Innovation: $k = 30$

$$A - bQ = u + sQ \tag{54}$$

$$\Rightarrow (b+s)Q = A - u$$

$$\Rightarrow Q_0 = \frac{A-u}{b+s} = \frac{170-50}{3} = 40$$

$$A - bQ = u - k + sQ \tag{55}$$

$$\Rightarrow Q_1 = \frac{A-u+k}{b+s} = \frac{170-50+30}{3} = 50$$

$$\Delta Q = Q_1 - Q_0 = \frac{k}{b+s} = \frac{30}{3} = 10 \tag{56}$$

$$\Delta P = -b \times \Delta Q = -k \times \frac{b}{b+s} \tag{57}$$

$$= -30 \times \frac{2}{3} = -20$$

- Movement along the demand curve (slope $-b$)

$$\rho = \frac{-\Delta P}{k} = \frac{b}{b+s} = \frac{2}{2+1} = \frac{2}{3} \tag{58}$$

$$k = \frac{b}{b+s}\, k + \frac{s}{b+s}\, k = 20 + 10 \tag{59}$$

- 20: the fall in the price
- 10: extra MC of 10 more units
- Every firm still sets $P = MC$

$$\rho = \frac{b}{b+s} = \frac{1/|\varepsilon_D|}{1/|\varepsilon_D| + 1/\varepsilon_S} \tag{60}$$
$$= \frac{\varepsilon_S}{\varepsilon_S + |\varepsilon_D|}$$

$$\rho = \frac{2.25}{2.25 + 1.125} = \frac{2.25}{3.375} = \frac{2}{3} \tag{61}$$

- $b = (P/Q)/|\varepsilon_D|$, $s = (P/Q)/\varepsilon_S$; P/Q cancels
- More elastic supply: more pass-through

$$\rho = \frac{b}{b+s} = \frac{1}{1+s/b} \xrightarrow{b\to\infty} 1 \tag{62}$$

$$b = 0: \quad \rho = \frac{0}{0+s} = 0 \tag{63}$$

$$s = 0: \quad \rho = \frac{b}{b+0} = 1 \tag{64}$$

- $b \to \infty$: vertical demand, Q fixed
- $b = 0$: horizontal demand, P fixed
- $s = 0$: flat supply, constant MC

$$\text{Tax £30:} \qquad u \longrightarrow u + 30 \tag{65}$$

$$\Delta P = +30 \times \frac{b}{b+s} \tag{66}$$

$$= 30 \times \frac{2}{3} = +20$$

$$\text{Producers} = +20 - 30 \tag{67}$$

$$= -10 \text{ per unit}$$

- Buyers pay 2/3 of the tax, producers 1/3
- A tax is the mirror image of the innovation
- Preview of Tutorial 3

$$MC_B(q) = \frac{dTC_B}{dq} = P_N + 50 + 10q \tag{68}$$

$$\begin{aligned} P_B &= P_N + 50 + 10q \qquad (69) \\ \Rightarrow 10q &= P_B - P_N - 50 \\ \Rightarrow q &= \frac{P_B - P_N - 50}{10} \end{aligned}$$

- $P_N + 50$ replaces the 50 of Exercise 1
- Valid for $P_B \geq P_N + 50$

$$Q_B = 10 \times \frac{P_B - P_N - 50}{10} \tag{70}$$
$$= P_B - P_N - 50$$

$$P_B = P_N + 50 + Q_B \tag{71}$$

- Intercept $P_N + 50$: nickel plus other cost
- Slope 1: rising other marginal cost

$$220 - Q_B = P_N + 50 + Q_B \tag{72}$$
$$\Rightarrow 2Q_B = 170 - P_N$$
$$\Rightarrow Q_B = 85 - 0.5P_N$$

$$0.5P_N = 85 - Q_B \tag{73}$$
$$\Rightarrow P_N = 170 - 2Q_B = 170 - 2Q_N$$

- The nickel demand curve of Exercise 2

$$A - bQ = P_N + c + gQ \tag{74}$$
$$\Rightarrow P_N = (A - c) - (b + g)Q$$

$$P_N = (220 - 50) - (1 + 1)Q \tag{75}$$
$$= 170 - 2Q$$

- $-b$: battery demand ($b = 1$)
- $-g$: rising other marginal cost ($g = 1$)
- $A - c = 220 - 50$: value of the first unit
- $A = 220$ here (battery demand), $A = 170$ in WE6 (nickel)

$$\Delta MC_N = -30 \longrightarrow \Delta P_N = -20 \quad (76)$$

$$\Delta P_N = -20 \longrightarrow \Delta MC_B = -20 \quad (77)$$

$$\Delta MC_B = -20 \longrightarrow \Delta P_B = -10 \quad (78)$$

- One unit of nickel per battery
- Battery supply shifts down by 20

$$\rho_N = \frac{20}{30} = \frac{2}{3} = \frac{b+g}{(b+g)+s} \tag{79}$$

$$\rho_B = \frac{10}{20} = \frac{1}{2} = \frac{b}{b+g} \tag{80}$$

$$\rho_{\text{total}} = \frac{10}{30} = \frac{1}{3} \tag{81}$$

- Nickel demand slope $b + g = 2$; $s = 1$
- Check Question 3.3: P_B from 180 to 170

$$\varepsilon_{D,B} = -1 \times \frac{180}{40} = -4.5 \tag{82}$$

$$\varepsilon_{S,B} = 1 \times \frac{180}{40} = 4.5 \tag{83}$$

$$\rho_B = \frac{4.5}{4.5 + 4.5} = \frac{1}{2} \tag{84}$$

- Demand $Q_B = 220 - P_B$: $dQ/dP = -1$
- Supply $Q_B = P_B - P_N - 50$: $dQ/dP = 1$

$$\varepsilon_{D,N} = -0.5 \times 2.25 = -1.125 \tag{85}$$

$$\varepsilon_{D,B} = -1 \times 4.5 = -4.5 \tag{86}$$

- Slope: $-0.5 = -1/(b+g)$ versus -1
- Scale P/Q: $90/40$ versus $180/40$
- Nickel is half of a battery's MC
- Derived demand is less elastic